

Conrad Borchers

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Education

Carnegie Mellon University

HUMAN-COMPUTER INTERACTION, PhD (ADVANCED TO CANDIDACY 09/2025)

HUMAN-COMPUTER INTERACTION, MS

University of Oxford

SOCIAL DATA SCIENCE, MSc

University of Tübingen

PSYCHOLOGY, BSc

Pittsburgh, PA, USA

08/2022 – present

08/2022 - 08/2024

Oxford, United Kingdom

10/2021 - 08/2022

Tübingen, Germany

10/2018 - 07/2021

Professional Experience

08/2022 - **Graduate Research Assistant**, Human-Computer Interaction Institute, Carnegie Mellon University
02/2020 - 10/2024 **Research Affiliate**, Lab of Christian Fischer, University of Tübingen
11/2021 - 07/2022 **Student Researcher and Team Lead**, Oxford Artificial Intelligence Society
11/2021 - 07/2022 **Graduate Research Assistant**, Saïd Business School, University of Oxford
05/2021 - 09/2021 **Research Intern**, CAHL Lab (Zachary A. Pardos), University of California, Berkeley
08/2020 - 01/2021 **Undergraduate Research Assistant**, Tübingen School of Education
02/2019 - 07/2020 **Undergraduate Research Assistant**, Leibniz Knowledge Media Research Center (IWM)

Awards, Fellowships, & Grants

PAPER AND THESIS DISTINCTIONS

2026 **Best Full Paper Nomination [F31]**, 16th International Learning Analytics and Knowledge Conference
2025 **Best Full Paper Award [F23]**, 20th European Conference on Technology Enhanced Learning
2025 **Best Long Paper Nomination [F19]**, 18th International Conference on Educational Data Mining
2025 **Best Short Paper Nomination [S11]**, 18th International Conference on Educational Data Mining
2025 **Best Short Paper Nomination [S9]**, 15h International Learning Analytics and Knowledge Conference
2024 **Best Student Paper Award [F13]**, International Conference on Quantitative Ethnography
2024 **Best Interactive Event Nominee [D1]**, 25h International Conference on Artificial Intelligence in Education
2024 **Best Student Paper Award [F9]**, 17th International Conference on Educational Data Mining
2024 **Best Short Paper Award [S7]**, 17th International Conference on Educational Data Mining
2024 **Best Paper Award (AIS) [F7]**, 26th Intl. Conference On Human-Computer Interaction (HCIII)
2023 **Best Full Paper Honorable Mention [F1]**, 13th International Learning Analytics and Knowledge Conference
2022 **OII Thesis Prize for Best MSc Dissertation**, Oxford Internet Institute
2021 **Best Poster Award [S1]**, 14th International Conference on Educational Data Mining

REVIEWER AWARDS

2025 **Best Reviewer Award**, 12th ACM Conference on Learning@Scale (L@S)
2025 **Outstanding Junior Reviewer**, Journal of Learning Analytics

EXTERNAL RESEARCH FUNDING

- 2026 **Large Grant: Improving Persistence through Adaptive Goal Setting in i-Ready - \$250,000.00**, Digital Promise, the Advancing Innovative Math Solutions (AIMS) Collaboratory, and Learning Data Insights. *Role: PI | Co-PIs: Aleven, Koedinger, Thomas*
- 2025 **Planning Grant: Improving Persistence through Adaptive Goal Setting in i-Ready - \$10,000.00**, Digital Promise, the Advancing Innovative Math Solutions (AIMS) Collaboratory, and Learning Data Insights. *Role: PI | Co-PIs: Aleven, Koedinger, Thomas*

INTERNAL RESEARCH FUNDING

- 2023 **GuSH Graduate Research Grant (Scaffolding Effects on Problem-Solving in Intelligent Tutoring Systems) [F4, F10, F13, F30, J10, J16, J19] - \$750**, Carnegie Mellon University

SELECTIVE FELLOWSHIPS, SCHOLARSHIPS, AND SUMMER SCHOOLS

- 2025 **Siebel Scholars Class of 2026 (\$35,000)**, Thomas and Stacey Siebel Foundation
- 2024 **Ethical AI in Education Fellow**, Learnest
- 2022 **LearnLab Summer School Participant (ITS Track)**, Carnegie Mellon University
- 2022 **Merck Innovation Cup '22 Participant**, Merck KGaA (Merck Group)
- 2021 **Learning Engineering Fellow - 2nd Place in Tools Competition (\$15,000)**, EPIC@Berkeley - UC Berkeley GSE & Schmidt Futures
- 2019/20 **Research Fellow**, European Federation of Psychology Students Associations (EFPSA)
- 2018 **BSc and MSc Scholarship**, German Academic Scholarship Foundation (Studienstiftung)

CONFERENCE FUNDING

- 2025 **Conference Attendance (EDM '25) - \$475**, NSF-SPLICE through the University of Pittsburgh
- 2025 **GSA Conference Funding (LAK '25) - \$750**, Carnegie Mellon University
- 2024 **GSA Conference Funding (LAK '24) - \$750**, Carnegie Mellon University
- 2024 **Full Conference Travel Support (AIED '24) - \$2,000**, Learnest
- 2023 **EDM2023 Travel Scholarship - \$1,250**, International Educational Data Mining Conference
- 2023 **Prof. Ramkumar Memorial Foundation Scholarship - \$100**, International Educational Data Mining Conference
- 2023 **GSA Conference Funding (LAK '23) - \$750**, Carnegie Mellon University

Teaching Experience

TEACHING ASSISTANT

- Fall 2025 **Large Language Models: Methods and Applications**, School of Computer Science, Carnegie Mellon University
- Fall 2023 **User-Centered Research and Evaluation**, School of Computer Science, Carnegie Mellon University
- Spring 2023 **Tools for Online Learning**, School of Computer Science, Carnegie Mellon University
- Spring 2021 **Cognitive Modeling**, Department of Computer Science, University of Tübingen, Germany

GUEST LECTURES

- Fall 2025 **CS 565: CS565**, Two Guest Lectures on Publication [W5] and [F20], respectively, hosted by Neil Heffernan, Worcester Polytechnic Institute
- Summer 2025 **HCII NSF-REU Summer Research Program Seminar**, Guest Lecture: "What Makes Good Research?," hosted by Laura Dabbish, Carnegie Mellon University
- Spring 2025 **Tools for Online Learning**, Guest Lecture on Socio-Emotional Learning, hosted by Steven J. Moore, Carnegie Mellon University
- Spring 2025 **CS 5890/6890: Topics in AI and Education**, Guest Lecture: "Enhancing Education with LLMs: Opportunities and Challenges," hosted by Yang Shi, Utah State University
- Fall 2024 **CS 568: AI in Adaptive Educational Technologies**, Guest Lecture on Publications [CS7] and [CF9], hosted by Neil Heffernan, Worcester Polytechnic Institute
- Spring 2024 **Tools for Online Learning**, Guest Lecture on Socio-Emotional Learning, hosted by Steven J. Moore, Carnegie Mellon University
- Summer 2023 **HCII NSF-REU Summer Research Program Seminar**, Guest Lecture: "What Makes Good Research?," hosted by Laura Dabbish, Carnegie Mellon University

WORKSHOPS

- Fall 2024 **Multimodal Machine Learning: An Introductory Workshop**, Hosted by Joshua Rosenberg at the University of Tennessee, Knoxville; GitHub: <https://tinyurl.com/utk-mmml>
- Fall 2022 **Sentence Embeddings for Twitter Research**, Introduction to Data Science Methods in Education, University of Tennessee, Knoxville; GitHub: <https://tinyurl.com/sembutk>

Academic Publications

Total citations: 844 (Google Scholar, 2026-04-04)

h-index: 16 (Google Scholar, 2026-04-04)

i10-index: 30 (Google Scholar, 2026-04-04)

Number of distinct co-authors on listed papers: 188

Full list of citations per paper: <https://scholar.google.com/citations?user=v5yiJVAAAAAJ&hl=en>

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Conference Acceptance Rates (2023-2025): LAK: 30%, EC-TEL: 27%, EDM: 26%, AIED: 21%, ISLS: 30%, LatS: 27%

Italicized author names indicate mentored students. An asterisk (*) indicates equal contribution.

JOURNAL ARTICLES

- J21 – Simon, S., Şener, B., Pifarré Turmo, M., Greisel, M., Čarapina, M., **Borchers, C.**, Shahrokhian, B., Tajik, E., Chernikova, O., Hatun Ataş, A., Hu, X., & Sitjà Martín, M. (in press). Toward a Taxonomy: Insights from the 2025 CSCL Paradigm Building Workshop. *International Journal of Computer-Supported Collaborative Learning*.
- J20 – **Borchers, C.**, Thomas, D. R., Lin, J., & Koedinger, K. R. (in press). Learning-Aware Reliability Estimation for Tutor Skill Assessment Using Large Language Models. *Journal of Learning Analytics*.
- J19 – Zambrano, A. F., Wei, Z., Zhang, J., Baker, R. S., Ocumpaugh, J. L., Barany, A., Liu, X., Zhou, Y., Paquette, L., Ginger, J., & **Borchers, C.** (2026). Data Plus Theory Equals Codebook: Leveraging LLMs for Human-AI Codebook Development. *Journal of Educational Data Mining*. <https://doi.org/10.5281/zenodo.18352290>
- J18 – **Borchers, C.**, Morina, F., Klein, L., & Fischer, C. (2025). Mapping the landscape of educational use of X (Twitter) in Germany: informal teacher learning in online communities of practice. *Journal of New Approaches in Educational Research*. <https://doi.org/10.1007/s44322-025-00047-8>

- J17 – **Borchers, C.** & Braesemann, F. (2025). The innovation dynamics of programming technologies. *Journal of the Royal Society Interface*. <https://doi.org/10.1098/rsif.2025.0166>
- J16 – **Borchers, C.**, Zhang, J., Fleischer, H., Schanze, S., Aleven, V. & Baker, R. S. (2025). Large Language Models Generalize SRL Prediction to New Languages Within But Not Between Domains. *Journal of Educational Data Mining*. <https://doi.org/10.5281/zenodo.17073680>
- J15 – **Borchers, C.** & Pardos, Z. A. (2025). Course Load Analytics Interventions on Higher Education Course Selection: Experimental Evidence. *Journal of Learning Analytics*. <https://doi.org/10.18608/jla.2025.8473>
- J14 – **Borchers, C.**, Fleischer, H., Schanze, S., Scheiter, K., & Aleven, V. (2025). High scaffolding of an unfamiliar strategy improves conceptual learning but reduces enjoyment compared to low scaffolding and strategy freedom. *Computers & Education*, 236, 105364. <https://doi.org/10.1016/j.compedu.2025.105364>
- J13 – **Borchers, C.**, Wang, Y., Hodge, E., & Rosenberg, J. M. (2025). Decoding Sentiment Signals: Lessons from the Political Reception of the Common Core and Next Generation Science Standards. *Educational Researcher*. <https://doi.org/10.3102/0013189X251336206>
- J12 – **Borchers, C.**, Xu, Y., & Pardos, Z. A. (2025). Workload Overload? Late Enrollment Leads to Course Dropout. *Journal of Educational Data Mining*. <https://doi.org/10.5281/zenodo.14907388>
- J11 – **Borchers, C.**, Darriet, C., Rosenberg, J., & López, F. (2025). Dual Intent in Dual-language Programs: Internet Data Mining of School District Communications. *TechTrends*. <https://doi.org/10.1007/s11528-025-01049-1>
- J10 – **Borchers, C.**, Fleischer, H., Yaron, D. J., McLaren, B. M., Scheiter, K., Aleven, V., & Schanze, S. (2025). Problem-Solving Strategies in Stoichiometry Across Two Intelligent Tutoring Systems: A Cross-National Study. *Journal of Science Education and Technology*. <https://doi.org/10.1007/s10956-024-10197-7>
- J9 – Zhang, L., Lin, J., Sabatini, J., **Borchers, C.**, Weitekamp, D., Cao, M., Hollander, J., Hu, X., & Graesser, A. C. (2025). Data Augmentation for Sparse Multidimensional Learning Performance Data Using Generative AI. *IEEE Transactions on Learning Technologies*. <https://doi.org/10.1109/TLT.2025.3526582>
- J8 – Pritchard, C., **Borchers, C.**, Rosenberg, J. M., Fox, A. K., & Stegenga, S. M. (2024). The datafication of student information on X (Twitter). *Computers and Education Open*, 7, 100197. <https://doi.org/10.1016/j.caeo.2024.100197>
- J7 – **Borchers, C.***, Rosenberg, J.*, & Swartzentruber, R. M.* (2023). Facebook Post Data: A Primer for Educational Research. *Educational Technology Research and Development*. <https://doi.org/10.1007/s11423-023-10269-2>
- J6 – **Borchers, C.***, Eder, T. F.*, Richter, J., Keutel, C., Huettig, F., & Scheiter, K. (2023). A Time Slice Analysis of Dentistry Students' Visual Search Strategies and Pupil Dilation during Diagnosing Radiographs. *PloS one*. <https://doi.org/10.1371/journal.pone.0283376>
- J5 – Pardos, Z. A., **Borchers, C.**, & Yu, R. (2023). Credit hours is not enough: Explaining undergraduate perceptions of course workload using LMS records. *The Internet and Higher Education*, 56, 100882. <https://doi.org/10.1016/j.iheduc.2022.100882>
- J4 – Rosenberg, J., **Borchers, C.**, Stegenga, S., Burchfield, M., Anderson, D., & Fischer, C. (2022). How educational institutions reveal students' personally identifiable information on Facebook. *Learning, Media and Technology*. Advanced Online Publication. <https://doi.org/10.1080/17439884.2022.2140672>
- J3 – Rosenberg, J., **Borchers, C.**, Burchfield, M., Anderson, D., Stegenga, S., & Fischer, C. (2022). Posts About Students on Facebook: A Data Ethics Perspective. *Educational Researcher*. Advanced Online Publication. <https://doi.org/10.3102/0013189X221120538>
- J2 – Rosenberg, J., Burchfield, M., **Borchers, C.**, Gibbons, B., Anderson, D., & Fischer, C. (2021). Social media and students' privacy: What schools and districts should know. *Phi Delta Kappan*, 103(2), 49-53. <https://doi.org/10.1177/00317217211051145>
- J1 – Rosenberg, J., **Borchers, C.**, Dyer, E., Anderson, D. & Fischer, C. (2021). Understanding Public Sentiment About Educational Reforms: The Next Generation Science Standards on Twitter. *AERA Open*, 7(1), 1-17. <https://doi.org/10.1177/23328584211024261>

PEER-REVIEWED CONFERENCE PROCEEDINGS (FULL PAPERS)

- F39 – **Borchers, C.**, Rodrigues, L., Torrezão da Costa, N., Xavier, C., & Mello, R. F. (in press). Understanding Teacher Revisions of Large Language Model-Generated Feedback. *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED 2026)*.

- F38 – **Borchers, C.**, Švábenský, V., Kafle, S. K., Tang, K. K., & Vykopal, J. (in press). Multimodal Analytics of Cybersecurity Crisis Preparation Exercises: What Predicts Success? *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED 2026)*.
- F37 – Thomas, D. R., **Borchers, C.**, Vanacore, K. P., Koedinger, K. R., & Kizilcec, R. F. (in press). Modernizing Ground Truth: Four Shifts Toward Improving Reliability and Validity in AI in Education. *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED 2026)*. Seoul, South Korea.
- F36 – Du, Y., **Borchers, C.**, & Cukurova, M. (in press). Benchmarking Educational LLMs with Analytics: A Case Study on Gender Bias in Feedback. *Proceedings of 27th International Conference on Artificial Intelligence in Education (AIED)*. Seoul, South Korea.
- F35 – Jin, Q., **Borchers, C.**, Gurung, A., Jackson, S., Agarwal, S., Wang, J., Yu, A., Maheshwary, P., & Aleven, V. (in press). Sticky Help, Fleeting Effects: Session-by-Session Analytics of Teacher Interventions in K-12 Classrooms. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785128>
- F34 – Švábenský, V.*, **Borchers, C.***, Fortuna, E. Cloude, E. B., & Gašević, D. (in press). Mapping 15 Years of Learning Analytics Research: Global Trends, Topics, and Collaboration. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785131>
- F33 – **Borchers, C.**, Deininger, H., & Pardos, Z. A. (in press). Toward Trait-Aware Learning Analytics. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785131>
- F32 – **Borchers, C.**, Gurung, A., Liu, Q., Thomas, D. R., Khalil, M., & Koedinger, K. R. (in press). Brief but Impactful: How Human Tutoring Interactions Shape Engagement in Online Learning. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785041>
- F31 – Liu, Q., Li, L., Švábenský, V., **Borchers, C.**, & Khalil, M. (in press). Measuring the Impact of Student Gaming Behaviors on Learner Modeling. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785036>
- F30 – Zhang, J., **Borchers, C.**, Wang, C., Kumar, V., Teffera, L., McLaren, B. M., & Baker, R. S. (in press). Understanding Gaming the System through Analyzing Self-Regulated Learning in Think-Aloud Protocols. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785025>
- F29 – Thomas, D. R., **Borchers, C.**, & Koedinger, K. R. (2025). Beyond Agreement: Rethinking Ground Truth in Educational AI Annotation. *NCME AIME-Con 2025*, Pittsburgh, PA, USA. <https://doi.org/10.48550/arXiv.2508.00143>
- F28 – Thomas, D. R., **Borchers, C.**, Bhushan, S., Gatz, E., Gupta, S., & Koedinger, K. R. (2025). LLM-Generated Feedback Supports Learning When Learners Choose to Use It. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03870-8_33
- F27 – Bhushan, S., Thomas, D. R., **Borchers, C.**, Raghuvanshi, I., Abboud, R., Gatz, E., Gupta, S., & Koedinger, K. R. (2025). Detecting LLM-Generated Short Answers and Effects on Learner Performance. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03870-8_4
- F26 – **Borchers, C.**, Nguyen, H. T., Carvalho, P. F., Koedinger, K. R., & Aleven, V. (2025). Goal Setting Engages More Caregivers Than Instructional Support in Online Math Homework. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03870-8_5
- F25 – **Borchers, C.**, Nagashima, T., Liu, P., Alibali, M. W., & Aleven, V. (2025). Is More Gamification Better? Evaluating Playful Interactions and Narratives for Algebra Learning. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03870-8_6
- F24 – Xia, M.*, Schmucker, R.*, **Borchers, C.***, & Aleven, V. (2025). Optimizing Mastery Learning by Fast-Forwarding Over-Practice Steps. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03870-8_37
- F23 – Teich, K., **Borchers, C.**, & Rummel, N. (2025). Using Log Data to Analyze the Impact of Adaptive Support on Self-Regulated Learning in Adult Online Education. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03870-8_31
- F22 – Simon, S., Tajik, E., **Borchers, C.**, Shahrokhian, B., Sankaranarayanan, S., Balzan, F., Strauss, S., Viswanathan, S. A., Atas, A. H., Carapina, M., Liang, L., & Celik, B. (2025). Comparing a Human's and a Multi-Agent System's Thematic Anal-

ysis: Assessing Qualitative Coding Consistency. *Proceedings of 26th International Conference on Artificial Intelligence in Education (AIED)*. Palermo, Italy. https://doi.org/10.1007/978-3-031-98420-4_5

- F21 – **Borchers, C.**, Houk, A., Aleven, V., & Koedinger, K. R. (2025). Engagement and Learning Benefits of Goal Setting with Rewards in Human-AI Tutoring. *Proceedings of 26th International Conference on Artificial Intelligence in Education (AIED)*. Palermo, Italy. https://doi.org/10.1007/978-3-031-98459-4_4
- F20 – **Borchers, C.** & Shou, T. (2025). Can Large Language Models Match Tutoring System Adaptivity? A Benchmarking Study. *Proceedings of 26th International Conference on Artificial Intelligence in Education (AIED)*. Palermo, Italy. https://doi.org/10.1007/978-3-031-98417-4_29
- F19 – Gurung, A., Lin, J., Huang, Z., **Borchers, C.**, Baker, R. S., Aleven, V., & Koedinger, K. R. (2025). Starting Seatwork Earlier as a Valid Measure of Student Engagement. *Proceedings of the 18th International Conference on Educational Data Mining (EDM)*. Palermo, Italy. <https://doi.org/10.5281/zenodo.15870183>
- F18 – Simon, S., **Borchers, C.**, Ataş, A. H., Tajik, E., Çelik, B., Čarapina, M., Liu, Y., Shahrokhian, B., Sankaranarayanan, S., Balzan, F., Molinari, G., Jagušt, T., & Liang, L. (2025). Shared Conceptual Ground in CSCL: Evidence From a Global Survey. In Oshima, J., Chen, B., Vogel, F., & Järvelä, J. (Eds.), *Proceedings of the 18th International Conference on Computer-Supported Collaborative Learning - CSCL 2025*. <https://doi.org/10.22318/csc12025.987431>
- F17 – Thomas, D. R., **Borchers, C.**, Kakarla, S., Lin, J., Bhushan, S., Guo, B., Gatz, E., & Koedinger, K. R. (2025). Do Tutors Learn from Equity Training and Can Generative AI Assess It? *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706531>
- F16 – Thomas, D. R., **Borchers, C.**, Kakarla, S., Lin, J., Bhushan, S., Guo, B., Gatz, E., & Koedinger, K. R. (2025). Does Multiple Choice Have a Future in the Age of Generative AI? A Posttest-Only RCT. *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706530>
- F15 – Venugopalan, D., Yan, Z., **Borchers, C.**, Lin, J., & Aleven, V. (2025). Combining Large Language Models with Tutoring System Intelligence: A Case Study in Caregiver Homework Support. *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706516>
- F14 – Švábenský, V., **Borchers, C.**, Cloude, E. B., & Shimada, A. (2025). Evaluating the Impact of Data Augmentation on Predictive Model Performance. *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706485>
- F13 – Zhang, J.* , **Borchers, C.*** , & Barany, A. (2024). Studying the Interplay of Self-Regulated Learning Cycles and Scaffolding Through Ordered Network Analysis Across Three Tutoring Systems. *Proceedings of the International Conference on Quantitative Ethnography (ICQE)*, Philadelphia, PA, USA. https://doi.org/10.1007/978-3-031-76335-9_17
- F12 – Yang, K. B.* , **Borchers, C.*** , Falhs, A.-C., Echeverria, V., Karumbaiah, S., Rummel, N., & Aleven, V. (2024). Leveraging Multimodal Classroom Data for Teacher Reflection: Teachers' Preferences, Practices, and Privacy Considerations. *Proceedings of the 19th European Conference on Technology Enhanced Learning (EC-TEL)*, Krems, Austria. https://doi.org/10.1007/978-3-031-72315-5_34
- F11 – Baucks, F.* , Schmucker, R.* , **Borchers, C.**, Pardos, Z. A., & Wiskott, L. (2024). Gaining Insights into Group-Level Course Difficulty via Differential Course Functioning. *Proceedings of the Tenth (2024) ACM Conference on Learning@Scale (L@S)*. Atlanta, GA, USA. <https://doi.org/10.1145/3657604.3662028>
- F10 – Zhang, J., **Borchers, C.**, Aleven, V., & Baker, R. S. (2024). Using Large Language Models to Detect Self-Regulated Learning in Think-Aloud Protocols. *Proceedings of the 17th International Conference on Educational Data Mining (EDM)*. Atlanta, GA, USA. <https://doi.org/10.5281/zenodo.12729790>
- F9 – **Borchers, C.**, Yang, K., Lin, J., Rummel, N., Koedinger, K. R., & Aleven, V. (2024). Combining Dialog Acts and Skill Modeling: What Chat Interactions Enhance Learning Rates During AI-Supported Peer Tutoring? *Proceedings of the 17th International Conference on Educational Data Mining (EDM)*. Atlanta, GA, USA. <https://doi.org/10.5281/zenodo.12729784>
- F8 – **Borchers, C.**, Liu, X., Lee, H. H., & Zhang, J. (2024). Ethical AIED and AIED Ethics: Toward Synergy Between AIED Research and Ethical Frameworks. *Proceedings of 25th International Conference on Artificial Intelligence in Education (AIED) – BlueSky Track*. Recife, Brazil. https://doi.org/10.1007/978-3-031-64315-6_2
- F7 – Zhang, L., Lin, J., **Borchers, C.**, Sabatini, J., Hollander, J., Cao, M., & Hu, X. (2024). Predicting Learning Performance with Large Language Models: A Study in Adult Literacy. *Proceedings of the 26th International Conference On Human-Computer Interaction (HCI)*. Washington, DC, USA. https://doi.org/10.1007/978-3-031-60609-0_24
- F6 – Nguyen, H. T., **Borchers, C.**, Xia, M., & Aleven, V. (2024). Designing Tools for Caregiver Involvement in Intelligent Tutoring Systems for Middle School Mathematics. *Proceedings of the 2024 Annual Meeting of the International Society of the*

- F5 – **Borchers, C.**, Wang, Y., Karumbaiah, S., Ashiq, M., Shaffer, D. W., & Aleven, V. (2024). Revealing Networks: Understanding Effective Teacher Practices in AI-Supported Classrooms using Transmodal Ordered Network Analysis. *Proceedings of the 14th International Learning Analytics and Knowledge Conference (LAK)*. Kyoto, Japan. ACM. <https://doi.org/10.1145/3636555.3636892>
- F4 – **Borchers, C.**, Zhang, J., Baker, R. S., & Aleven, V. (2024). Using Think-Aloud Data to Understand Relations between Self-Regulation Cycle Characteristics and Student Performance in Intelligent Tutoring Systems. *Proceedings of the 14th International Learning Analytics and Knowledge Conference (LAK)*. Kyoto, Japan. ACM. <https://doi.org/10.1145/3636555.3636911>
- F3 – **Borchers, C.**, Carvalho, P. F., Xia, M., Liu, P., Koedinger, K. R., & Aleven, V. (2023). What Makes Problem-Solving Practice Effective? Comparing Paper and AI Tutoring. *Proceedings of the 18th European Conference on Technology Enhanced Learning (EC-TEL)*. Aveiro, Portugal. https://doi.org/10.1007/978-3-031-42682-7_4
- F2 – Karumbaiah, S., **Borchers, C.**, Shou, T., Falhs, A.-C., Liu, C., Nagashima, T., Rummel, N., & Aleven, V. (2023). A Spatiotemporal Analysis of Teacher Practices in Supporting Student Learning and Engagement in an AI-enabled Classroom. *Proceedings of the 24th International Conference on Artificial Intelligence in Education (AIED)*. Tokyo, Japan. https://doi.org/10.1007/978-3-031-36272-9_37
- F1 – **Borchers, C.** & Pardos, Z. A. (2023). Insights into undergraduate pathways using course load analytics. *Proceedings of the 13th International Learning Analytics and Knowledge Conference (LAK)*. Arlington, TX, USA. ACM. <https://doi.org/10.1145/3576050.3576081>

PEER-REVIEWED CONFERENCE PROCEEDINGS (SHORT/POSTER PAPERS)

- S27 – **Borchers, C.**, Zhang, J., & Gurung, A. (in press). Representation Learning to Study Temporal Dynamics in Tutorial Scaffolding. *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED 2026)*. Seoul, South Korea.
- S26 – Lyu, Q*, **Borchers, C.***, Xia, M., Xiao, K., Carvalho, P. F., Koedinger, K. R., & Aleven, V. (in press). Evaluating a Data-Driven Redesign Process for Intelligent Tutoring Systems. *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED 2026)*. Seoul, South Korea.
- S25 – Huang, X., **Borchers, C.**, Zhang, J., & Lajoie, S. P. (in press). Physiological and Semantic Patterns in Medical Teams Using an Intelligent Tutoring System. *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED 2026)*. Seoul, South Korea.
- S24 – Tajik, E., Shahrokhian, B., **Borchers, C.**, Simon, S., Keramati, A., Pal, S., & Sankaranarayanan, S. (in press). Disagreement as Data: Reasoning Trace Analytics in Multi-Agent Systems. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785101>
- S23 – Zhang, J., **Borchers, C.**, Cohn, C., Srivastava, N., Snyder, C., Guo, S., Ashwin, T. S., Mohammed, N., Noh, H., & Biswas, G. (in press). Using Large Language Models to Detect Socially Shared Regulation of Learning in Collaborative Learning. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785083>
- S22 – **Borchers, C.**, Patel, M., Lee, S., & Botelho, A. F. (in press). Disentangling Learning from Judgment: Representation Learning for Open Response Analytics. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785042>
- S21 – Thomas, D. R., **Borchers, C.**, Lin, J., Kakarla, S., Bhushan, S., Gatz, E., Gupta, S., Abboud, R., & Koedinger, K. R. (2025). Leveraging LLMs to Assess Tutor Moves in Real-Life Dialogues: A Feasibility Study. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03873-9_35
- S20 – Falhs, A.-C., **Borchers, C.**, Echeverria, V., Yang, K., Rummel, N., & Aleven, V. (2025). How Expertise Levels Shape Preferences and Reflection Needs: Towards AI Reflection Systems for Teacher Empowerment. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03873-9_11
- S19 – Fleischer, H., **Borchers, C.**, Schanze, S., & Aleven, V. (2025). Error Classification in Stoichiometry Tutoring Systems with Different Levels of Scaffolding: Comparing Rule-Based Classification and Machine Learning. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03873-9_12

- S18 – **Borchers, C.**, Sheel, S., Pai, A., Shah, S., & Pardos, Z. A. (2025). Designing the Course Load Analytics Platform. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03873-9_10
- S17 – Chen, E., Tang, X., Xi, A., Lin, C., **Borchers, C.**, Gupta, S., Lin, J., & Koedinger, K. R. (2025). VTutor for High-Impact Tutoring at Scale: Managing Engagement and Real-Time Multi-Screen Monitoring with P2P Connections. *Proceedings of the 12th ACM Conference on Learning at Scale (L@S)*, Palermo, Italy. <https://doi.org/10.1145/3698205.3733948>
- S16 – Fleischer, H., Noglik, A., **Borchers, C.**, & Schanze, S. (2025). Does Student Learning Rate Depend on Feedback Type and Prior Knowledge? *Proceedings of the 18th International Conference on Educational Data Mining (EDM 2025)*. Palermo, Italy. <https://doi.org/10.5281/zenodo.15870227>
- S15 – Jin, Q., **Borchers, C.**, Fanscali, S., & Aleven, V. (2025). Who to Help? A Time-Slice Analysis of K-12 Teachers' Decisions in Classes with AI-Supported Tutoring. *Proceedings of the 18th International Conference on Educational Data Mining (EDM 2025)*. Palermo, Italy. <https://doi.org/10.5281/zenodo.15870229>
- S14 – **Borchers, C.**, Nguyen, H. T., Carvalho, P. F., Koedinger, K. R., & Aleven, V. (2025). Involving Parents in Tutoring Systems to Increase Content Confidence: A Design Probe Study *Proceedings of 26th International Conference on Artificial Intelligence in Education (AIED)*. Palermo, Italy. https://doi.org/10.1007/978-3-031-98465-5_14
- S13 – **Borchers, C.**, Peng, C., Lyu, Q., Carvalho, P. F., Koedinger, K. R., & Aleven, V. (2025). Student Perceptions of Adaptive Goal Setting Recommendations: A Design Prototyping Study. *Proceedings of 26th International Conference on Artificial Intelligence in Education (AIED)*. Palermo, Italy. https://doi.org/10.1007/978-3-031-98462-4_31
- S12 – Thomas, D. R., **Borchers, C.**, Bhushan, S., Kakarla, S., Houk, A., Abboud, R., Gatz, E., Gupta, S., & Koedinger, K. R. (2025). Improving Open-Response Assessment with LearnLM. *Proceedings of 26th International Conference on Artificial Intelligence in Education (AIED)*. Palermo, Italy. https://doi.org/10.1007/978-3-031-98462-4_28
- S11 – **Borchers, C.** (2025). Toward Sufficient Statistical Power in Algorithmic Bias Assessment: A Test for ABROCA. *Proceedings of the 18th International Conference on Educational Data Mining (EDM)*. Palermo, Italy. <https://doi.org/10.5281/zenodo.15870256>
- S10 – Butler, R., **Borchers, C.**, Asher, M., Lee, Y., Karnataki, S., Dangi, S., Athreya, S., Stamper, J., Ogan, A., & Carvalho, P. (2025). Does the Doer Effect Generalize to Non-WEIRD Populations? Toward Analytics in Radio and Phone-Based Learning. *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706505>
- S9 – **Borchers, C.** & Baker, R. S. (2025). ABROCA Distributions for Algorithmic Bias Assessment: Considerations Around Interpretation. *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706498>
- S8 – **Borchers, C.***, Ooge, J.*, Peng, C., & Aleven, V. (2025). How Learner Control and Explainable Learning Analytics About Skill Mastery Shape Student Desires to Finish and Avoid Loss in Tutored Practice. *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706484>
- S7 – **Borchers, C.**, Xu, Y., & Pardos, Z. A. (2024). Are You an Early Dropper or Late Shopper? Mining Enrollment Transaction Data to Study Procrastination in Higher Education. *Proceedings of the 17th International Conference on Educational Data Mining (EDM)*. Atlanta, GA, USA. <https://doi.org/10.5281/zenodo.12729852>
- S6 – Peng, C., **Borchers, C.**, & Aleven, V. (2024). Designing Homework Support Tools for Middle School Mathematics Using Intelligent Tutoring Systems. *Proceedings of the 2024 Annual Meeting of the International Society of the Learning Sciences (ISLS)*. Buffalo, NY, USA. <https://doi.org/10.22318/icls2024.989202>
- S5 – Shou, T., **Borchers, C.**, Karumbaiah, S., & Aleven, V. (2023). Optimizing Parameters for Accurate Position Data Mining in Diverse Classrooms Layouts. *Proceedings of the 16th International Conference on Educational Data Mining (EDM)*. Bengaluru, India. <https://doi.org/10.5281/zenodo.8115685>
- S4 – **Borchers, C.**, Klein, L., Johnson, H., & Fischer, C. (2023). Timing Matters: Inferring Educational Twitter Community Switching from Membership Characteristics. *Proceedings of the 16th International Conference on Educational Data Mining (EDM)*. Bengaluru, India. <https://doi.org/10.5281/zenodo.8115752>
- S3 – Karumbaiah, S.*, **Borchers, C.***, Falhs, A.-C.*, Holstein, K., Rummel, N., & Aleven, V. (2023). Teacher Noticing and Student Learning in Human-AI Partnered Classrooms: A Multimodal Analysis. *Proceedings of the 2023 Annual Meeting of the International Society of the Learning Sciences (ISLS)*. Montréal, Canada. <https://doi.org/10.22318/icls2023.151200>

- S2 – **Borchers, C.**, Rosenberg, J., Gibbons, B., Burchfield, M., & Fischer, C. (2021). To Scale or Not to Scale: Comparing Popular Sentiment Analysis Dictionaries on Educational Twitter Data. *Proceedings of the 14th International Conference on Educational Data Mining (EDM)*. Paris, France.
- S1 – Burchfield, M., Rosenberg, J., **Borchers, C.**, Thomas, T., Gibbons, B., & Fischer, C. (2021). Are Violations of Student Privacy “Quick and Easy”? Investigating the Privacy of Students’ Images and Names in the Context of K-12 Educational Institution’s Posts on Facebook. *Proceedings of the 14th International Conference on Educational Data Mining (EDM)*. Paris, France.

PEER-REVIEWED WORKSHOP AND DOCTORAL CONSORTIUM PUBLICATIONS

- W9 – **Borchers, C.**, Koedinger, K. R., & Aleven, V. (2025). Intelligent Support for Practice Goal Setting to Enhance Learning. *Doctoral Consortium of the 26th International Conference on Artificial Intelligence in Education (AIED)*. https://doi.org/10.1007/978-3-031-99261-2_43
- W8 – **Borchers, C.***, Tian, X.*, Boyer, K. E., Israel, M. (2025). Combining Log Data and Collaborative Dialogue Features to Predict Project Quality in Middle School AI Education. *9th Educational Data Mining in Computer Science Education (CSEDM) Workshop*.
- W7 – Sankaranarayanan, S., **Borchers, C.**, Simon, S., Tajik, E., Atas, A. H., Celik, B., Balzan, F., & Shahrokhian, B. (2025). Automating Thematic Analysis with Multi-Agent LLM Systems. *From Data to Discovery: LLMs for Qualitative Analysis in Education at LAK25*.
- W6 – **Borchers, C.**, Thomas, D. R., Lin, J., Abboud, R., & Koedinger, K. R. (2025). Augmenting Human-Annotated Training Data with Large Language Model Generation and Distillation in Open-Response Assessment. *Proceedings of the Second Workshop on Generative AI for Learning Analytics (GenAI-LA) at LAK25*.
- W5 – Kakarla, S., **Borchers, C.**, Thomas, D. R., Bhushan, S., & Koedinger, K. R. (2025). Comparing Few-Shot Prompting of GPT-4 LLMs with BERT Classifiers for Open-Response Assessment in Tutor Equity Training. *Proceedings of the Innovation and Responsibility in AI-Supported Education (iRAISE) Workshop at AAAI25*.
- W4 – Aleven, V., **Borchers, C.**, Huang, Y., Nagashima, T., McLaren, B., Carvalho, P., Popescu, O., Sewall, J., & Koedinger, K. (2024). An Integrated Platform for Studying Learning with Intelligent Tutoring Systems: CTAT+TutorShop. *Proceedings of the Fifth Annual Workshop on Learning@Scale 2024: A/B Testing and Platform-Enabled Learning Research*.
- W3 – Zhang, L., Lin, J., **Borchers, C.**, Cao, M., & Hu, X. (2024). 3DG: A Framework for Using Generative AI for Handling Sparse Learner Performance Data From Intelligent Tutoring Systems. *Proceedings of the First Workshop on Generative AI for Learning Analytics (GenAI-LA) at LAK24*.
- W2 – Han, Z. F., Lin, J., Gurung, A., Thomas, D. R., Chen, E., **Borchers, C.**, Gupta, S., & Koedinger, K. R. (2024). Improving Assessment of Tutoring Practices using Retrieval-Augmented Generation. *Proceedings of the AAAI2024 Workshop on AI for Education - Bridging Innovation and Responsibility*.
- W1 – **Borchers, C.***, Gala, D. S.*, Gilbert, B.*, Oravkin, E., Bounsi, W., Asano, Y. M., & Kirk, H. R. (2022). Looking for a Handsome Carpenter! Debiasing GPT-3 Job Advertisements. *Proceedings of the 4th Workshop on Gender Bias in Natural Language Processing (NAACL)*.

PEER-REVIEWED CONFERENCE DEMO PUBLICATIONS

- D3 – Chen, E., Tang, X., Xi, A., Lin, C., **Borchers, C.**, Gupta, S., Lin, J., & Koedinger, K. R. (2025). Demo of VTutor for High-Impact Tutoring at Scale: A Real-Time Multi-Screen Tutor Support System with P2P Connections. *Proceedings of the 12th ACM Conference on Learning at Scale (L@S)*, Palermo, Italy.
- D2 – Yang, K., Zhao, Y., **Borchers, C.**, Fang, Y., Kim, J., Butt, A., Popescu, O., Gupte, S., Chiu, A., Shen, P., Rummel, N., & Aleven, V. (2025). Reflecto: A Teacher Reflection Tool Leveraging Multimodal Learning and Teaching Analytics. *26th International Conference on Artificial Intelligence in Education (AIED)*.
- D1 – Rao, J., **Borchers, C.**, & Lin, J. (2024). Coursera-REC: Explainable MOOCs Course Recommendation using RAG-facilitated LLMs. *25th International Conference on Artificial Intelligence in Education (AIED)*.

Presentations & Invited Talks

CONFERENCE PRESENTATIONS

- P9 – Fleischer, H., **Borchers, C.**, Schanze, S., & Aleven, V. (2024, September). Fehlerklassifizierung beim Tutor-Gestützten Lösen von Stöchiometrieaufgaben [Error Classification in Tutor-Supported Stoichiometry Problem Solving]. Presentation at the 2024 Annual Meeting of the Gesellschaft für Didaktik der Chemie und Physik (GDChP), Bochum, Germany.
- P8 – Rosenberg, J. M., Pritchard, C., **Borchers, C.**, Burchfield, M., Fischer, C., Fox, A. K., & Stegenga, S. (2024, March). Data ethics in the open: Navigating student information risks in educational social media. Presentation at the Data Science & Social Research, 4th International Conference. <https://dssr2024.unina.it/index.html>
- P7 – Gold, G., **Borchers, C.**, & Carvalho P. F. (2024, March). Students' Academic Performance and Goal Orientation Relate to Initial Knowledge but Not Learning Rate. Poster presented at the 14th International Learning Analytics and Knowledge Conference (LAK), Kyoto, Japan.
- P6 – Fischer, C., Morina, F., **Borchers, C.**, & Klein, L. (2023, August). Online Communities of Practice on Twitter: A Case Study of the German Educational Twittersphere. Paper presented at the 20th Biennial EARLI Conference, Thessaloniki, Greece.
- P5 – **Borchers, C.**, & Braesemann, F.. Correlation networks predict success of digital technologies (2022, November). Paper presented at the 11th International Conference on Complex Networks and their Applications, Palermo, Italy.
- P4 – Fischer, C., Morina, F., **Borchers, C.**, Klein, L., & Eich, M. (2022, April). Informal Professional Development in Online Communities of Practice at Scale: Mapping Nationwide Educational Twitterspheres. Paper presented at the 2022 Annual Meeting of the American Educational Research Association, San Diego, CA.
- P3 – **Borchers, C.**, Eder, T., & Scheiter, K. (2021, September). How Effective is Reflective search? A Time Slice Analysis of Dentistry Students' Visual Search Strategies and Pupil Dilation during Diagnosing Radiographs. Paper presented at the 2nd Workshop on Mental Effort, Virtual Conference.
- P2 – Rosenberg, J., **Borchers, C.**, Gibbons, B., Dyer, E., Anderson, D., & Fischer, C. (2021, April). Don't worry, be happy? A sentiment analysis of the #NGSSchat Twitter community. Paper presented at the 2021 Annual Meeting of the American Educational Research Association, Virtual Conference.
- P1 – Fischer, C., Morina, F., **Borchers, C.**, & Klein, L. (2021, March). Using Big Data to Support Educational Change: Twitter as a Platform for Informal Professional Development. Paper presented at the 2021 Annual Meeting of the Leibniz Education Research Network, Hannover, Germany.

INVITED TALKS

- T18 – **Borchers, C.** (2026, March). Learning Analytics for Persistence: Designing Effort-Aware AI Systems in K-12 and Higher Education. College of Education and Human Development Texas A&M University.
- T17 – **Borchers, C.** (2026, February). Modeling Student Effort: Data Science for Persistence, Workload, and Learning. Department of Data Science, New Jersey Institute of Technology.
- T16 – **Borchers, C.** (2026, February). Artificial Intelligence for Modeling Effort, Workload, and Persistence: From K-12 Classrooms to Higher Education Institutions. Peabody College and College of Connected Computing, Vanderbilt University.
- T15 – **Borchers, C.** (2026, February). Data Science Approaches to Modeling and Supporting Student Persistence and Workload. School of Data Science, University of Virginia.
- T14 – **Borchers, C.** (2026, February). Computational Models of Effort, Workload, and Skill Acquisition in Educational Contexts. Department of Psychology, University of Illinois Urbana-Champaign.
- T13 – **Borchers, C.** (2025, December). Using Log Data to Analyze the Impact of Adaptive Support on Self-Regulated Learning in Adult Online Education [F23]. Collaboration Call, AIMS (Advancing Innovative Math Solutions) Collaboratory.
- T12 – **Borchers, C.** (2025, December). Goal Support in Hybrid Tutoring: Impact, Scale, and Lessons Learned. Collaboration Call, Learning Engineering Virtual Institute.
- T11 – **Borchers, C.** (2025, November). Toward Effort-Sensitive AI: Modeling and Supporting Student Persistence Across Contexts. GSE Colloquium Series in Education Data Science, Stanford University.
- T10 – **Borchers, C.** (2025, November). Effort-Sensitive AI for Learning: Closing Persistence Gaps Through Adaptive Design. Invited talk at the Digital Learning Lab (Mark Warschauer), University of California, Irvine.
- T9 – **Borchers, C.** (2025, October). Shaping Self-Regulated Learners in the LLM Era: Evidence, Models, and Design. *Information Science Colloquium*. Ann S. Bowers College of Computing and Information Science, Cornell University.
- T8 – **Borchers, C.** (2025, October). Effort-Sensitive AI for Learning: Measuring and Shaping Persistence. *Learning Sciences & Learning Technologies: Research Directions for the Next Decade*. École polytechnique fédérale de Lausanne (EPFL).

- T7 – **Borchers, C.** (2025, May). Combining LLMs with Tutoring System Intelligence: A Case Study in Caregiver Homework Support. LearnLab Cooperate Partners Summit.
- T6 – **Borchers, C.** (2025, April). Designing Tools for Caregiver Involvement in Intelligent Tutoring Systems. GSA Scholar to Scholar Carnegie Mellon University. **GSA Academic Affairs Member Award 2024/25.**
- T5 – **Borchers, C.** (2024, November). LLMs in Educational Research: Recent Trends and Three Future Directions. Interdisciplinary Education Seminar at the University of Tennessee, Knoxville.
- T4 – **Borchers, C.** (2024, September). To Improve Education with LLMs, Do More Than Just LLMs. Learning Analytics (AMA Series; M.S.Ed. Online), University of Pennsylvania GSE
- T3 – **Borchers, C.** (2024, May). Envisioning LLMs as Tools for Educational Research and Practice Rather than Instructional Replacements. Invited Talk at the ALTTAI Seminar at the University of Memphis.
- T2 – **Borchers, C.** (2022, October). Debiasing GPT-3 Job Ads. The Data Skeptic Podcast hosted by Kyle Polich.
- T1 – **Borchers, C.** (2022, September). Leveraging data science to improve educational processes: From K-12 student privacy on social media to higher education student workload. Invited talk at the Department of Theory & Practice in Teacher Education, University of Tennessee, Knoxville.

INVITED PANELS

- IP4 – Generative AI to streamline institutional and system workflows (2026, March). OECD Digital Education Outlook Conference.
- IP3 – Virtual Literacy Panel Series – Technology and Literacy: Building Skills for Learning and Careers (2025, October). Carnegie Library of Pittsburgh hosted by Robert Longo.
- IP2 – Early Career Researchers and AI (2025, October). Empowering Learners in the Age of AI (ELAI) Global Conference 2025.
- IP1 – Early Career Panel (2025, July). 26th International Conference on Artificial Intelligence in Education (AIED).

Mentoring

Spring 2026	Ishan Gandhi , Undergraduate Research Assistant, Carnegie Mellon University
Spring 2026	Carina Zheng , Independent Undergraduate Research, Carnegie Mellon University
Spring 2026	Meenakshi Nair , Independent Undergraduate Research, Carnegie Mellon University
Spring 2026	Sebastien Nyiri , Undergraduate Independent Study, Carnegie Mellon University
2025/26	Kent Grass , Research Assistant, PLUS Tutoring, Carnegie Mellon University
Fall 2025	Aidan Plummer , Undergraduate Independent Study, Carnegie Mellon University
Fall 2025	Ian Youn , Undergraduate Independent Study, Carnegie Mellon University
Summer 2025	Manit Patel , SURA, School of Computer Science, Carnegie Mellon University
Summer 2025	Hana Benko , SURA, School of Computer Science, Carnegie Mellon University
Summer 2025	Aarna Chowdhary , SURA, School of Computer Science, Carnegie Mellon University
Summer 2025	Arush Khare , SURA, School of Computer Science, Carnegie Mellon University
Summer 2025	Haley Noh , NSF REU, Carnegie Mellon University
Summer 2025	Karen Xiao , NSF REU, Carnegie Mellon University
Summer 2025	Shi (Eric) Qiu , NSF REU, Carnegie Mellon University
Spring 2025	Vardaan Gupta , Undergraduate Independent Study, Carnegie Mellon University
Spring 2025	Kevin Tang , Independent Undergraduate Research, Carnegie Mellon University
2024/25	Katherine Zhao , SCS SURA and Lab Intern, Carnegie Mellon University
2024/25	Siyu (Dorian) Pan , Lab Intern, Carnegie Mellon University
Summer 2024	Hrugved Anantrao Pawar , Summer Lab Intern, Carnegie Mellon University
Summer 2024	Kloe Chen , Summer Lab Intern, Carnegie Mellon University
Summer 2024	Doris Gao , Summer Lab Intern, Carnegie Mellon University
Summer 2024	Zoey Yan , NSF REU, Carnegie Mellon University

Summer 2024	Devika Venikopalan , SURA, School of Computer Science, Carnegie Mellon University
Fall 2023	Snigdha Tiwari , Research Assistant, Carnegie Mellon University
Fall 2023	Dhruvisha Mondhe , METALS Independent Study, Carnegie Mellon University
2023/24/25	Kennion Gubler , NSF REU and Independent Study, Carnegie Mellon University
Summer 2023	Ha Tien Nguyen , NSF REU, Carnegie Mellon University
Summer 2023	Cindy Peng , NSF REU, Carnegie Mellon University
Spring 2023	Jialin Ye , MHCI Independent Study, Carnegie Mellon University
Spring 2023	Avani Guduri , Undergraduate Independent Study, Carnegie Mellon University
2022/24/25	Tianze (Steven) Shou , Independent Study, Carnegie Mellon University

Service to the Academic Community

WORKSHOPS ORGANIZED

2026	A CACL Taxonomy Building Workshop , Martin Greisel, Burcu Sener, Sebastian Simon, Conrad Borchers, Elham Tajik and Bahar Shahrokhian	<i>ISLS'26</i>
2025	CACL Paradigm Building Workshop , Conrad Borchers, Francesco Balzan, Berkan Celik, Bahar Shahrokhian, Amine Hatun Ataş, Mia Čarapina, Sebastian Simon, Elham Tajik, Gaëlle Molinari and Carol Kwai Kuen Chan	<i>ISLS'25</i>

PEER REVIEW

2026	Telematics and Informatics Reports , Ad-hoc Reviewer
2026	International Journal of Human-Computer Studies , Ad-hoc Reviewer
2026	Assessing Writing , Ad-hoc Reviewer
2026	Array , Ad-hoc Reviewer
2025	AIS Transactions on Human-Computer Interaction , Ad-hoc Reviewer
2025-2026	Discover Education , Ad-hoc Reviewer
2025	PeerJ Computer Science , Ad-hoc Reviewer
2025	AIED Encyclopedia – Chapter Review , Ad-hoc Reviewer
2025	Entertainment Computing , Ad-hoc Reviewer
2025	European Journal of Education , Ad-hoc Reviewer
2025-2026	Computers & Education: Artificial Intelligence , Ad-hoc Reviewer
2025	Acta Psychologica , Ad-hoc Reviewer
2025-2026	Journal of Computer Assisted Learning , Ad-hoc Reviewer
2025	PLOS One , Ad-hoc Reviewer
2025	Information Processing and Management , Ad-hoc Reviewer
2025	Computers in Human Behavior Reports , Ad-hoc Reviewer
2025	IEEE Transactions on Knowledge and Data Engineering , Ad-hoc Reviewer
2025-2026	Computers & Education , Ad-hoc Reviewer
2025-2026	Social Sciences & Humanities Open , Ad-hoc Reviewer
2025	Studies in Educational Evaluation , Ad-hoc Reviewer
2025	Computers in Human Behavior: Artificial Humans , Ad-hoc Reviewer
2025	Cogent Education , Ad-hoc Reviewer
2025	Journal of Science Education and Technology , Ad-hoc Reviewer
2025	International Journal of Adolescence and Youth , Ad-hoc Reviewer
2025	BMC Medical Education , Ad-hoc Reviewer
2025	International Journal of Human-Computer Interaction , Ad-hoc Reviewer
2025-2026	Journal of Educational Data Mining , Ad-hoc Reviewer
2024-2025	IEEE Transactions on Learning Technologies , Ad-hoc Reviewer

2024 **Frontiers in Artificial Intelligence**, Ad-hoc Reviewer
 2024 **Springer Scientific Reports**, Ad-hoc Reviewer
 2024-2025 **British Journal of Educational Technology**, Ad-hoc Reviewer
 2024 **Frontiers in Education**, Ad-hoc Reviewer
 2024-2026 **Journal of Learning Analytics**, Ad-hoc Reviewer
 2024 **Journal of Infrastructure, Policy and Development**, Ad-hoc Reviewer
 2024-2026 **Teaching and Teacher Education**, Ad-hoc Reviewer
 2024 **American Journal of Education**, Ad-hoc Reviewer
 2025 **ICIS**, Conference Proceedings
 2024 **RecSys**, Conference Proceedings
 2024-2026 **L@S**, Conference Proceedings
 2024 **CSCW**, Conference Proceedings
 2024-2026 **EDM**, Conference Proceedings
 2024-2026 **AIED**, Conference Proceedings
 2024-2026 **CHI**, Conference Proceedings
 2024-2026 **LAK**, Conference Proceedings
 2024 **AAAI**, Workshop Track
 2023-2026 **ISLS**, Conference Proceedings
 2022-2024 **AERA**, Annual Meeting Proceedings
 2023 **NeurIPS**, Workshop Track

EDITORIAL POSITIONS

2025 **Junior Editorial Officer**, International Journal of Artificial Intelligence in Education

COMMITTEE MEMBERSHIPS

2026 **Executive Committee Student Member**, Society for Learning Analytics Research (SoLAR)
 2025 **IJAIED Track Chair**, International Conference on Artificial Intelligence in Education (AIED)
 2025 **Research Track Junior Advisory Board Member**, International Conference on Artificial Intelligence in Education (AIED)
 2023-2026 **Graduate Student Assembly Representative**, Carnegie Mellon University
 2023-2025 **NSF REU Summer Intern Admissions Committee**, Carnegie Mellon University

POLICY & ADVISORY CONTRIBUTIONS

2026 **OECD Digital Education Outlook 2026**, Pardos, Z. & Borchers, C., Chapter 11: *AI in Institutional Workflows: Learning from Higher Education to Unlock New Affordances for Education Systems and Institutions*, in *OECD Digital Education Outlook 2026: Exploring Effective Uses of Generative AI in Education* (OECD Publishing).
Institute of Education Sciences (IES), Co-author, Response to RFI on Education Research
 2025 **Priorities** (Lead: J. Roschelle; with S. Ritter, A. Closser, C. Borchers). Submitted recommendations on leveraging large-scale research infrastructure through SEERNet.

VOLUNTEERING

2023-2024 **EDM**, Student Volunteer
 2023-2024 **ISLS**, Student Volunteer